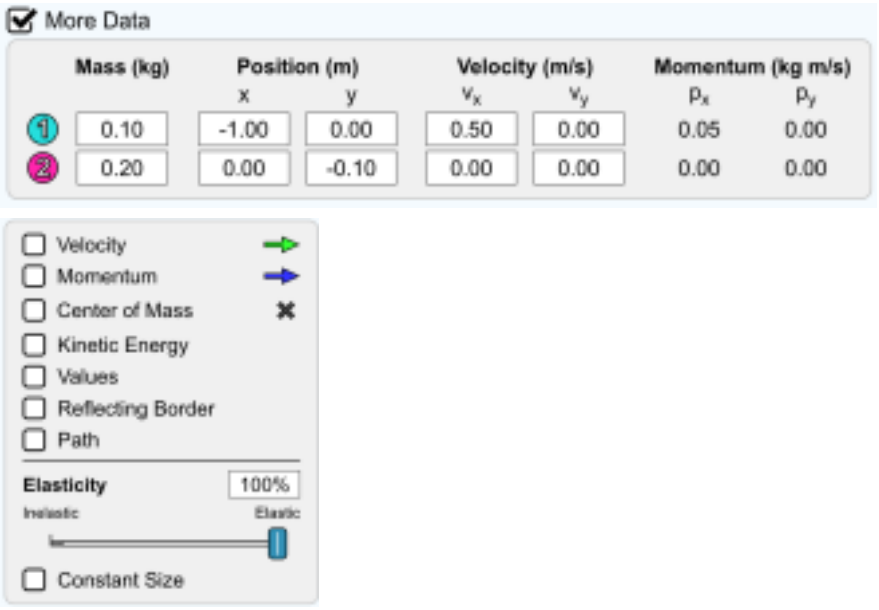
SPH4U1-01 LAB 2. COLLISIONS LAB

Name: _____Arunjan Jeyapalan_____ Mark: ____/36

1. Goal: Verify conservation of momentum and energy in elastic collisions.
2. Go to https://phet.colorado.edu/sims/html/collision-lab/latest/collision-lab_all.html and select "Explore 2D".
3. Set the initial conditions:

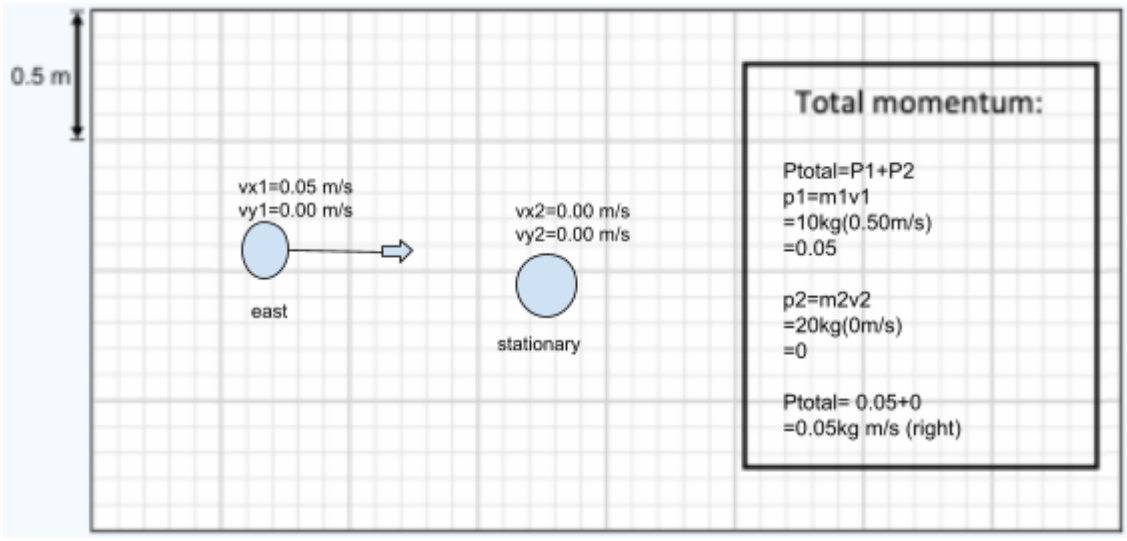


4. Hit and observe the motion before and after the collision. Use to pause and to reset the simulation.
5. Complete the table below, by writing the magnitudes and directions of all momentum vectors: (8 marks)

Ball	$\vec{p}_1$	$\vec{p}_2$	$\vec{p}_1'$	$\vec{p}_2'$
1	0.05 kg m/s	0.00 kg m/s	0.00 kg m/s	0.03 kg m/s
2	0.00 kg m/s	0.00 kg m/s	0.05 kg m/s	-0.03 kg m/s

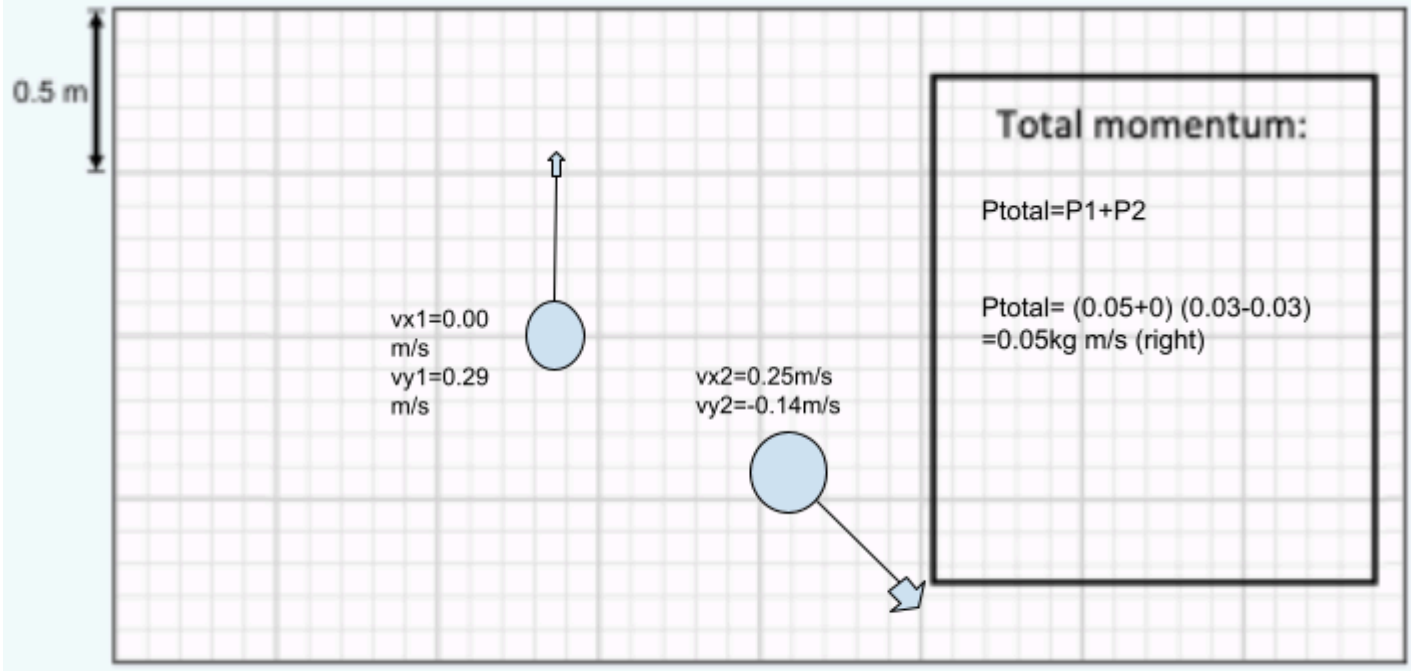
Note: use $m \cdot v$ for momentum.

6. In the diagram below, draw and label the momentum of each ball ($p1$ and $p2$) and the total momentum (P) before the collision: (Note: use vector addition (and show your work) to determine the total momentum.)



(5 marks)

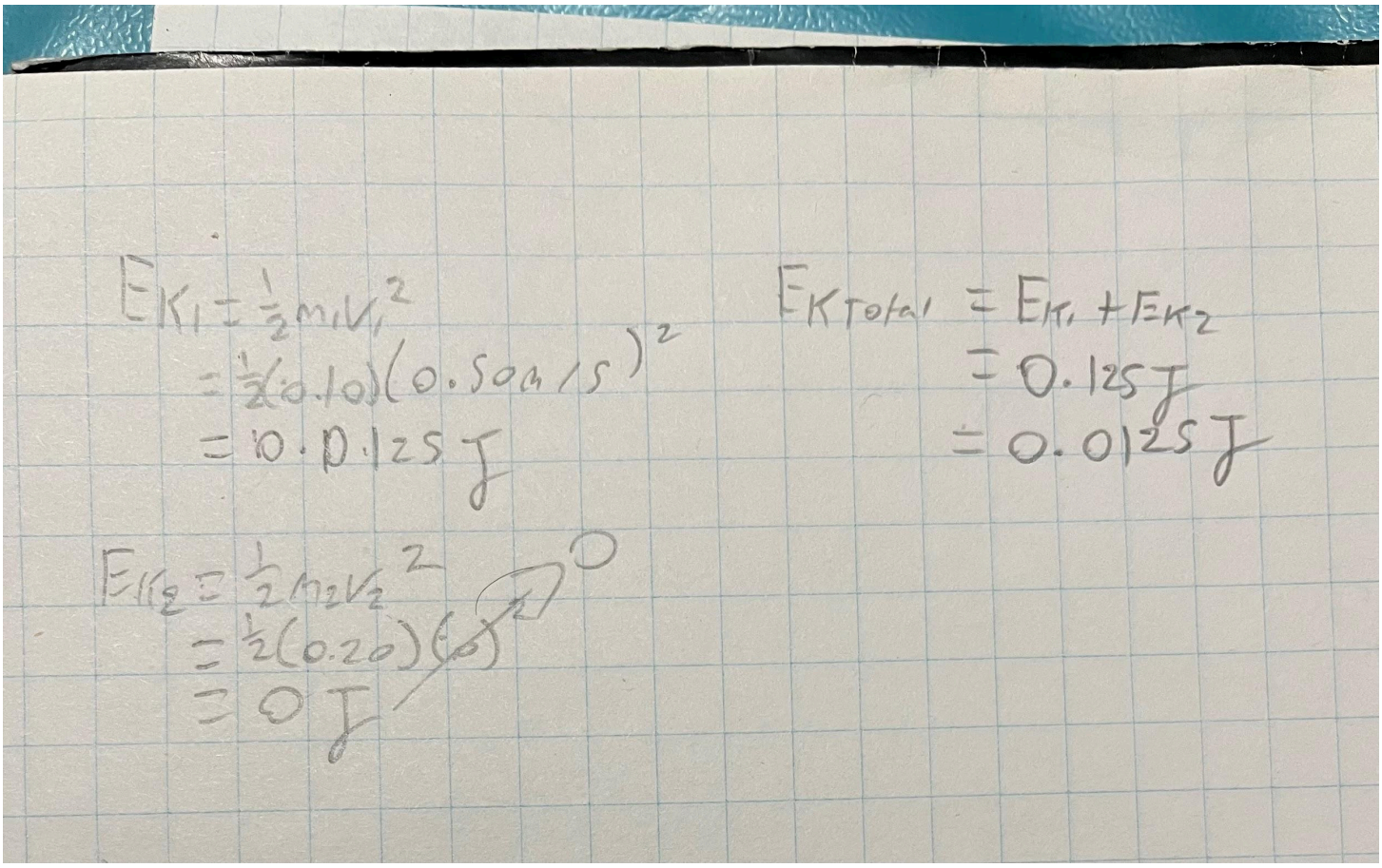
7. In the diagram below, draw and label the momentum of each ball ($\vec{p}_1$ and $\vec{p}_2$) and the total momentum ($\vec{p}$) after the collision: (Note: use vector addition (and show your work) to determine the total momentum.) (5 marks)



8. Was the total momentum conserved? Explain. (4 marks)

As the conservation of momentum rule states since there were no external forces in play, the total momentum was conserved since initial and final total momentum are both (0.05,0).

_ 9. Calculate the total kinetic energy before the collision (show your work): (3 marks)



Answer: the total kinetic energy before the collision was ___0.0125_____ J. (1 mark) 10. Calculate the total kinetic energy <u>after</u> the collision (show your work): (3 marks)

$$E_{K_1}' = \frac{1}{2} m_1 (v_1')^2$$
$$= \frac{1}{2}(0.10)(0.30)$$
$$= 0.0045\ J$$

$$E_{K_2}' = \frac{1}{2} m_2 (v_2')^2$$
$$= \frac{1}{2}(0.20)\left((0.25)^2 + (-0.15)^2\right)\ m/s^2$$
$$= \frac{1}{2}(0.20)(0.085)$$
$$= 0.0085\ J$$

$$E_{K\ TOTAL}' = E_{K_1}' + E_{K_2}'$$
$$= 0.0045\ J + 0.0085\ J$$
$$= 0.013\ J$$

Answer: the total kinetic energy after the collision was ___0.013_____ J. (1 mark)

11. Was the total kinetic energy conserved? Explain. (4 marks)

The total kinetic energy was conserved since the initial energy was 0.0125J, and the final energy was 0.013J. The values are extremely similar and the difference is negligible. Since both answers are approximately the same the total kinetic energy was conserved.

12. Conclusion: (2 marks)

Since the beginning and final momentum of the collision were both (0.05, 0), the total momentum was conserved. With the beginning kinetic energy at 0.0125, and the final value being 0.013J, the kinetic energy was also conserved since they are approximately the same.